FOSSDAQ

ai-4x-4to20mA

Build Manual
4-channel 4–20 mA current input card on TS-35 DIN rail

Project: FOSSDAQ

Module: ai-4x-4to20mA

Author: Sergej Tsoy

Supervision: Prof. Dr. Peter Stein, HTWG Konstanz

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1 Basic data

> *The `ai-4x-4to20mA` card measures the loop current of up to **four** 4–20 mA sensors. Four INA219 modules communicate with an Arduino Nano R4 over I²C. The controller sends the measured current in integer microamperes to a Raspberry Pi via USB. The Raspberry Pi / Node-RED performs the sensor-specific conversion, for example from current to flow in l/min.*

Property	Value
Function	4 analog current inputs for 4–20 mA loops
Controller	Arduino Nano R4
Acquisition	4 INA219 modules; one shunt per sensor loop
Interface	I²C on A4 / SDA and A5 / SCL; addresses 0x40, 0x41, 0x44, 0x45
Bus supply	24 V DC; common ground; input and pass-through
Sensor supply	Shared adjustable LM2596S output; 5 V shown in the schematic; verify the exact sensor rating
INA219 logic supply	Nano R4 5 V pin; common GND
Transfer	Integer current in µA; USB serial at 9600 baud
Example	12.00 mA is transmitted as 12000
Mounting	Printed enclosure for a TS-35 DIN rail
Isolation	No galvanic isolation between channels, sensors and controller

Module supply

The converter output is adjustable between **5 V and 21 V** for this build. The assembled prototype operates at 5 V according to the builder’s test. Select the output voltage according to the connected sensors and verify it with a multimeter before connection. The converter supplies the sensor L+ rail and Nano VIN. Connect INA219 VCC to the Nano’s regulated 5 V rail.

2 Components

Parts required for one card. See `../bom/BOM.csv` for the complete bill of materials, including all individual wire lengths, order codes and product links. Wire lengths are combined below for the build manual.

Qty	Part	Order code	Supplier	Purpose
1	Arduino Nano R4	ARD NANO R4	Reichelt	I²C acquisition and USB serial communication
2	Double-sided perfboard, 80 × 20 mm, 2.54 mm pitch	LR-DS-28	Reichelt	Sensor terminal carrier and cut strip for the bus terminal
1	JOY-IT LM2596S DC/DC step-down converter with display	DEBO DCDC DOWN 5	Reichelt	Adjustable module supply
4	INA219 current-sensor breakout	DEBO SENS POWER	Reichelt	One current measurement per sensor loop
4	6-pole spring terminal, 5 mm pitch	AST 025-06	Reichelt	Sensor terminals
1	4-pole spring terminal, 5 mm pitch	AST 025-04	Reichelt	24 V bus input and pass-through
1	USB-A to USB-C data cable, 2 m	EAN 5705730336140	Reichelt	Data link and Nano power supply
≈ 6 m	H05V-K stranded wire: 4 m at 0.75 mm² and 2 m at 0.5 mm²; red, orange, black, blue, brown, green and purple	See BOM.csv	Conrad	Power and I²C wiring; individual colors, lengths and EANs listed in BOM.csv
1 set	Self-tapping PA Phillips head screws: 4 × M1.2 × 6 mm and 24 × M2.3 × 6 mm	B0G6SWF81Z	Amazon	Fasten electronics and lid
min. 150 g	PLA / PETG filament	-	free choice	Printed enclosure parts

Tools: soldering iron, solder, fine saw, side cutter, wire stripper, multimeter and suitable screwdrivers. Deburr cut perfboard edges and printed parts. A current-limited DC supply and a suitable loop-current source or compatible sensor are needed for commissioning.

3 Circuit

3.1 Schematic

The card consists of a 24 V bus terminal, an LM2596S converter, an Arduino Nano R4, four INA219 modules and four sensor terminals. The sensor currents are measured in the return paths. Controller, converter and INA219 logic grounds share a common reference. Editable circuit sources are in `hardware/kicad/ai-4x-4to20mA.kicad_sch`.

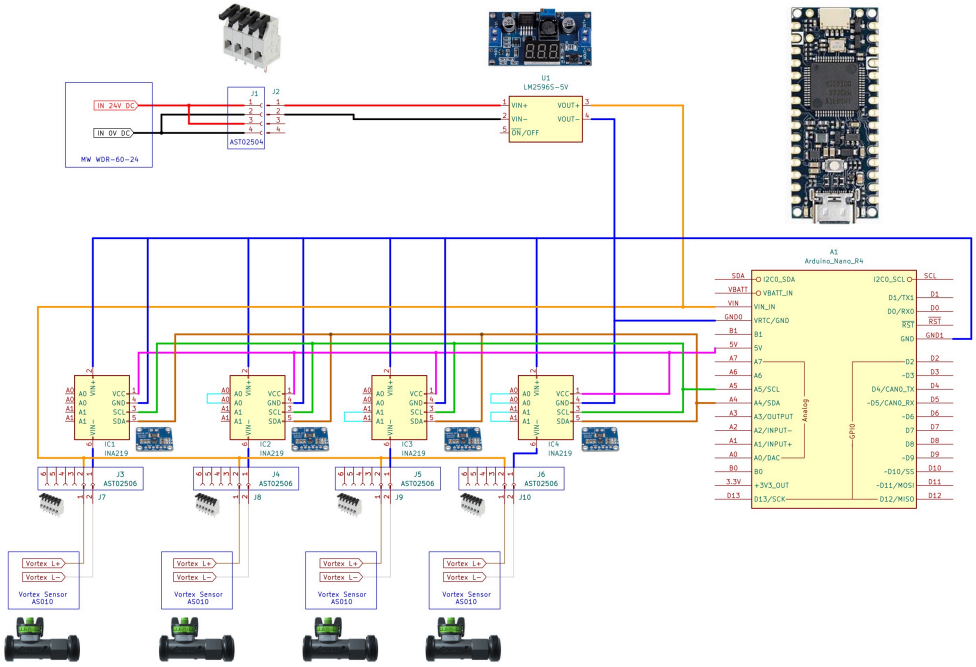


Figure 1: Schematic of the ai-4x-4to20mA card.

Section	Function
Bus terminal J1 / J2	Receives 24 V and GND; passes both rails to the next module
DC/DC converter U1	Supplies the sensor L+ rail and Nano VIN
INA219 IC1–IC4	Measure each loop return current across an individual shunt; communicate over I²C
Nano R4 A1	Averages current readings and returns µA on request over USB
Sensor terminals J3–J6	Connect L+ to the converter rail and L− to the assigned INA219 VIN−

Shunt polarity in the supplied schematic

The schematic connects each INA219 **VIN+ to common GND** and **VIN− to the sensor L− return**. Normal loop current therefore produces a negative signed reading. The supplied firmware averages ten samples and sends the absolute value in µA. This documents the supplied polarity; the absolute value alone cannot detect reverse current or incorrect loop wiring. Never bridge the sensor L− directly to GND: that would bypass its measuring shunt.

3.2 Current budget

The converter feeds the sensor loops and the Nano VIN input. The Nano's 5 V rail supplies the INA219 logic.

Item	Current / check
Four sensor loops	16–80 mA at nominal 4–20 mA per loop
Four INA219 ICs	Up to approximately 4 mA combined IC supply current; breakout components add load
Nano R4 and converter display	Determine actual consumption for the operating configuration
Converter and Nano regulator	Check the specific module rating, voltage, wiring and temperature under load

The loop current and electronics consumption are different budget items. For a VIN-powered Nano, convert its input power to converter output current at the selected voltage; do not simply add a 5 V current figure at another voltage. For USB power, include the Nano and INA219 logic in the USB budget. Allow for sensor fault/overrange current according to the sensor datasheet. No continuous 2 A rating is assumed for an unidentified LM2596S breakout.

4 Sensor connections and conversion

4.1 Terminal allocation

Each 6-pole terminal is assigned to **one** loop. The schematic uses poles 1–2; poles 3–6 have no sensor connection shown. Check the physical orientation against the schematic and enclosure labels before connecting a sensor. The photos show two rows of terminals; do not infer pole numbering from left-to-right order in a photograph.

Terminal pole	Function
1	L−: sensor return to the individual INA219 VIN−; not a direct GND connection
2	L+: sensor supply from the converter
3–6	Unused in the supplied schematic

Connect each sensor terminal to the INA219 module assigned below. Use the same channel numbers for the enclosure labels.

Housing channel	USB index	Terminal / mating part	INA219	I²C address
1	0	J3 / J7	IC1 / ina0	0x40
2	1	J4 / J8	IC2 / ina1	0x41
3	2	J6 / J10	IC4 / ina3	0x45
4	3	J5 / J9	IC3 / ina2	0x44

4.2 AS007 vortex sensor / 4–20 mA conversion

The autosen AS007 provides a 4–20 mA flow output. Connect sensor pin 1 (L+) to the terminal's L+ supply and pin 4 (OUT1, flow) to its L− measuring return. The optional temperature output on pin 2 is not used here. The supplied circuit illustration labels its example sensors AS010; the conversion below applies specifically to the AS007.

Property	Value
Sensor	autosen AS007, DN10, G3/4A
Flow measuring range	1.8–32 l/min
Flow output	4–20 mA; sensor pin 4 / OUT1
Firmware acquisition	10 samples per request, with 5 ms delay per sample
Transmitted value	Integer current in µA: `x = round(abs(mean(I_mA)) * 1000)`
Water conversion	`Q_l_min = 2 * (I_mA - 4)`
Pi / Node-RED conversion	`Q_l_min = (x - 4000) / 500`

For water, the 4–20 mA output corresponds to a 0–32 l/min analog scale. The lower specified measuring limit is 1.8 l/min; it is not the 4 mA scaling endpoint. With a received value of 12000 µA:

$$I = \frac{12000}{1000} = 12\ \mathrm{mA}$$

$$Q = 2 \times (12 - 4) = 16\ \mathrm{l/min}$$

For glycol mixtures with kinematic viscosity u above 1.8 cSt and below 14 cSt, the manufacturer gives the corrected AS007 relationship:

```
Q_l_min = (x - 4000) / 500 - 0.40 * u + 0.40
```

The corresponding response threshold is `Q_min = u + 0.8` in l/min. Use the viscosity for the actual mixture and temperature. Below the response threshold, the stated measurement accuracy does not apply.

Source: autosen AS006–AS011 operating instructions.

5 Enclosure and 3D printing

The enclosure contains the terminal carriers, four INA219 modules, controller and converter. The cover closes the side; a separate printed latch pin secures the card to the DIN rail. Editable CAD files belong in `hardware/cad/`; print files belong in `hardware/print/`.

Part	FreeCAD file	Function
Main body	`enclosure_body+slate.FCStd`	Holds the electronics and provides the terminal openings and DIN-rail mounting
Lid	`enclosure_lid.FCStd`	Closes the enclosure and is fastened with screws
Latch pin	`enclosure_pin.FCStd`	Locks the module onto the DIN rail

5.1 Main body

The main body holds the terminal carriers, four INA219 modules, the LM2596S converter and the Arduino Nano R4. The printed part has openings for the 24 V bus terminal, the four sensor terminals and the USB-C connector. The DIN-rail latch pin is inserted into the main body and secures the module to the rail.

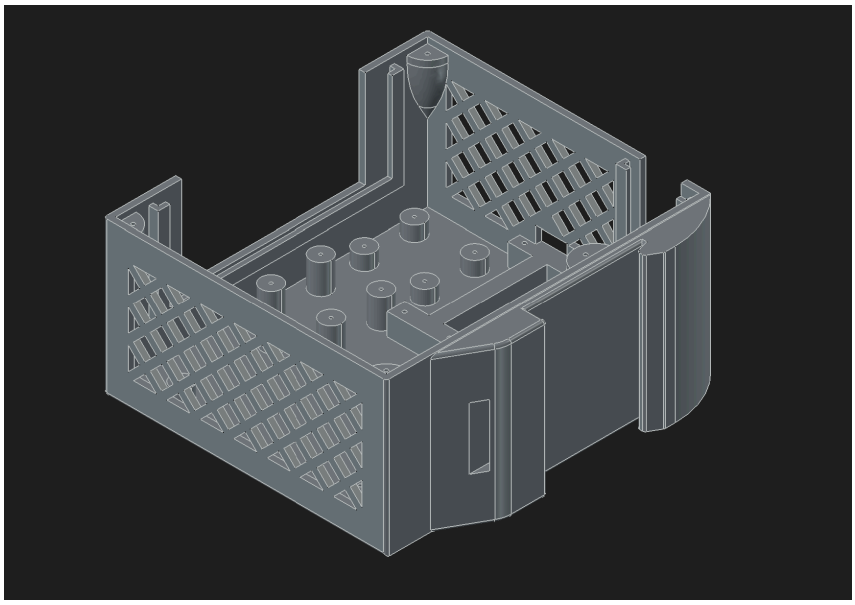


Figure 2: Open main enclosure

5.2 Lid

The lid closes the enclosure and is fastened with four self-tapping screws.

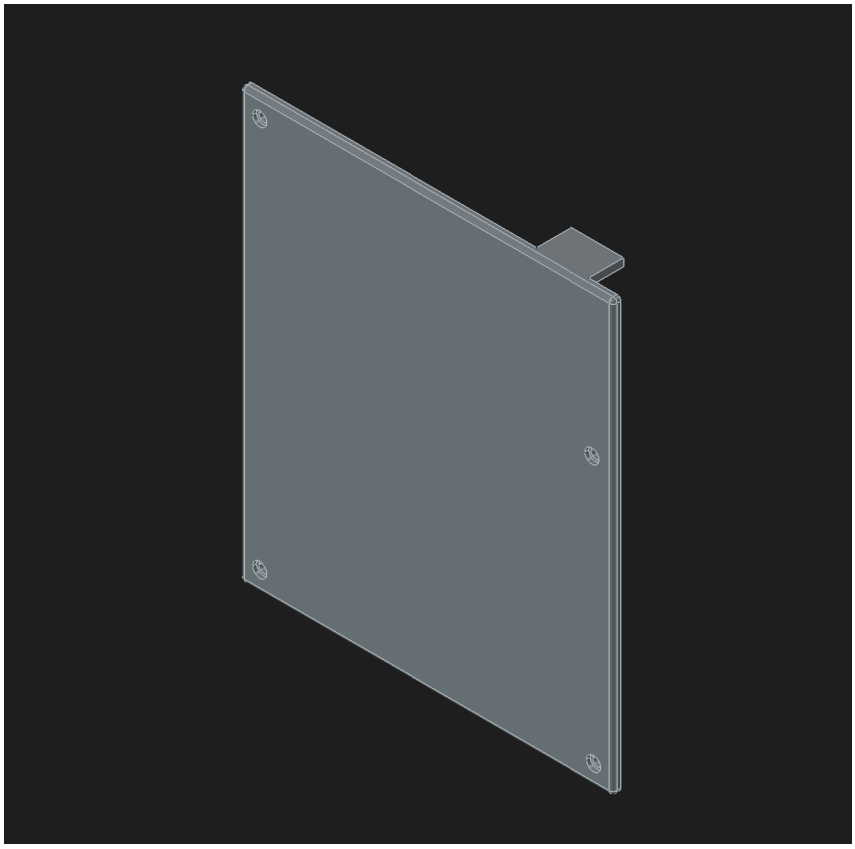


Figure 3: Side cover

5.3 DIN-rail latch pin

The pin is a separately printed springy element of the same DIN-rail latch type as the pi-base pin. Pushing the enclosure onto the rail makes it snap in; to release it, pull the pin back with a flat screwdriver or similar tool.

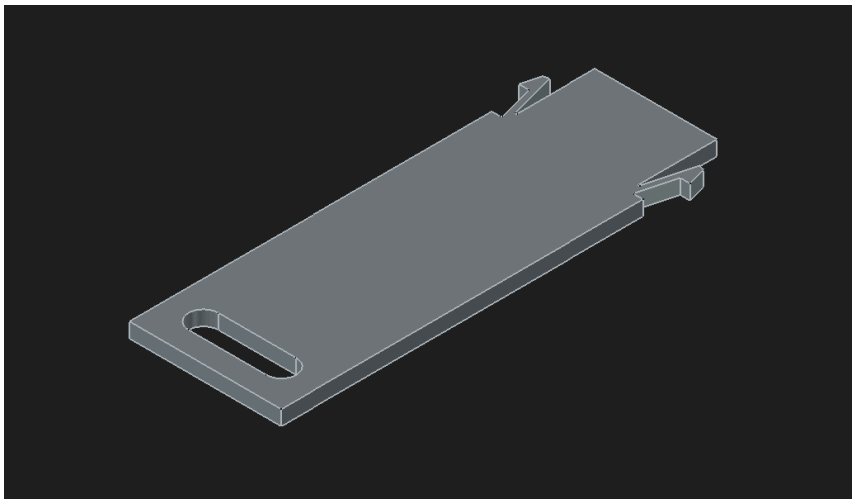


Figure 4: DIN-rail latch pin

5.4 3D printing

The print process follows the same approach as the pi-base module. The temperatures below are starting values; adjust them to the filament manufacturer and printer.

Parameter	Value
Filament	PLA or alternatively PETG
Nozzle / bed (PLA)	210 °C / 60 °C
Nozzle / bed (PETG)	230 °C / 80 °C
Supports	not required
Attention	thin walls of the slide-in slot - check the first layer carefully

6 24 V bus terminal assembly

The bus input sits on a narrow strip cut from the 80 × 20 mm perfboard. Two terminal poles receive 24 V and GND, and the second pair provides pass-through for the next module. Bridge only matching supply potentials.

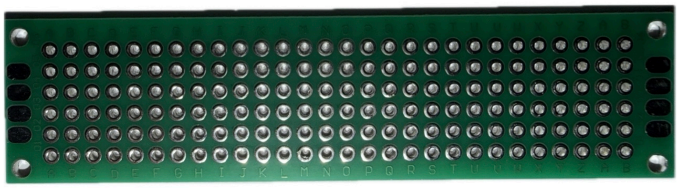


Figure 5: Raw double-sided perfboard, 80 × 20 mm.

1. **Cut and deburr the strip.** Use the marked cut lines as a guide and check the fit in the enclosure before soldering.
2. **Mount the 4-pole terminal.** Solder the AST 025-04 to the strip and connect the two 24 V poles to each other and the two ground poles to each other.
3. **Check continuity.** Verify the pass-through connections and confirm there is no short circuit between 24 V and GND.
4. **Install the carrier.** Slide the completed bus-terminal carrier into its guide.

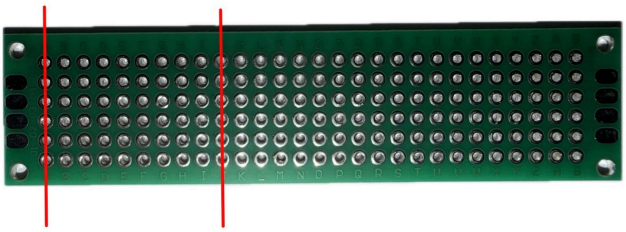


Figure 6: Marked cuts for the bus-terminal carrier.

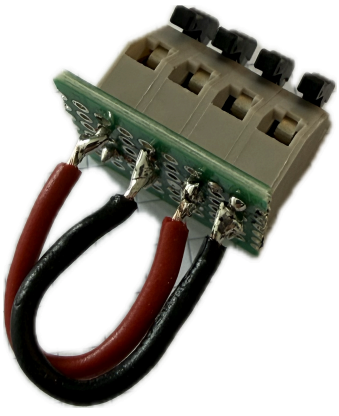


Figure 7: Bus terminal with pass-through bridges, solder side.

7 Sensor terminal carrier board

The carriers hold four 6-pole terminals; one terminal is assigned to each sensor loop. Use the carrier positions shown in the enclosure and photographs.

> **Allow clearance for the enclosure wall**
>
> Leave approximately 2 mm between the terminal bodies and the carrier board where the enclosure design requires it. Test the fit before completing the solder joints: the enclosure wall must fit into this gap.

1. **Place the four terminals.** Maintain the required orientation and spacing.
2. **Solder from the rear.** Keep the four individual L− terminals separate.
3. **Bridge the sensor supply rail.** Join only the four pole-2 L+ connections to the converter output. Pole-1 L− connections must not be bridged together or connected directly to common GND.
4. **Prepare four return wires.** Connect each pole-1 L− to VIN− of its assigned INA219. Connect each INA219 VIN+ to common GND as in the supplied schematic.
5. **Prepare the logic connections.** Connect VCC and GND of the four INA219 modules to the Nano’s 5 V and common GND; share SDA and SCL separately.
6. **Check the completed board.** Verify the supply rail, isolated loop returns and individual shunt paths before inserting the carriers.

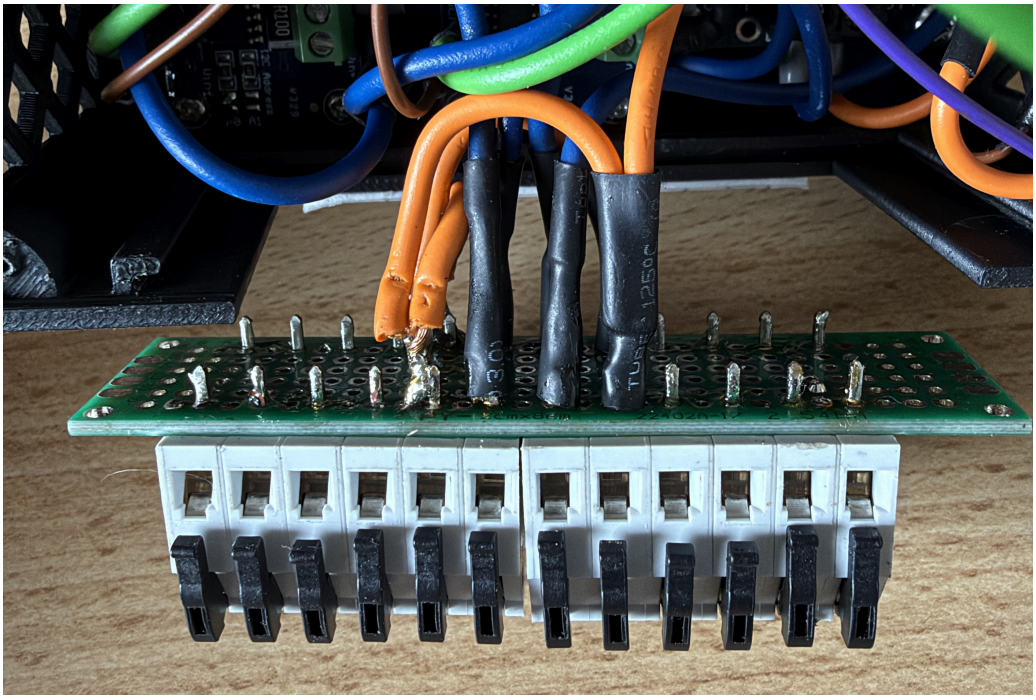


Figure 8: Sensor-terminal carrier: shared L+ supply and separate shunt return wiring.

8 Arduino Nano R4 wiring

Connect the four INA219 modules in parallel on the Nano R4′s primary I²C bus. Their addresses must be distinct. Configure the breakout address jumpers with power off; check the actual board’s jumper layout before soldering.

INA219 module	Address	A0	A1
IC1	0x40	GND	GND
IC2	0x41	VCC	GND
IC3	0x44	GND	VCC
IC4	0x45	VCC	VCC

Pin / connection	Purpose
A4 / SDA	Common INA219 SDA; brown in the schematic
A5 / SCL	Common INA219 SCL; green in the schematic
5 V	Regulated supply to all INA219 VCC pins; violet in the schematic
GND0, GND1	Common converter, INA219 logic and measuring-return reference
VIN	Supply from the converter; 5–21 V adjustment range for this build
USB-C	Serial link to the Pi; USB can power the Nano when VIN is disconnected

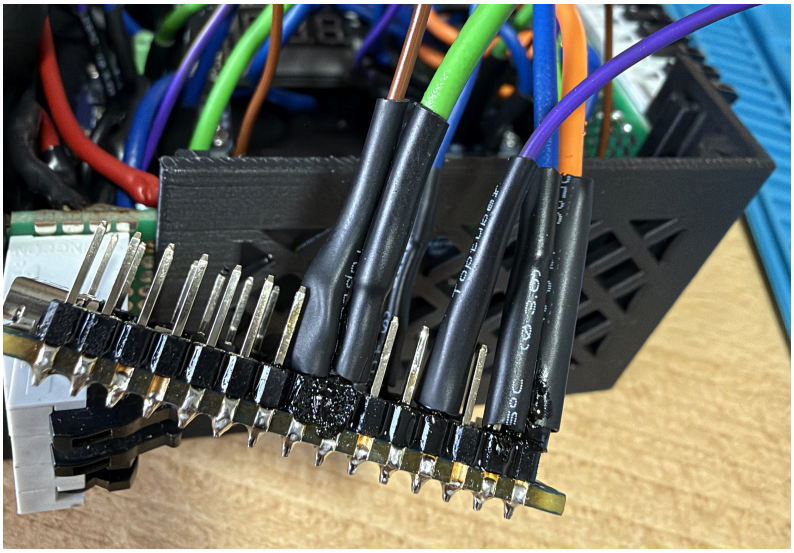


Figure 9: Prototype wiring at the Arduino Nano R4 headers; insulated soldered connections.

Match connections to the printed pin labels rather than wire color alone. Use A4/A5 for this 5 V I²C wiring; the Nano R4′s Qwiic interface is a separate 3.3 V interface. Keep SDA/SCL wiring short and check the combined pull-ups when using several breakout boards. Verify all four I²C addresses before sensor testing.

9 Mechanical assembly and converter setup

1. **Deburr the enclosure.** Check the carrier slots and terminal openings.
2. **Mount the converter and four INA219 modules.** Use the printed supports and appropriate screws. Keep the potentiometer accessible; avoid contact between the modules, solder joints and unrelated wiring.
3. **Connect the converter input.** Connect VIN+ to the 24 V bus and VIN− to common 0 V, observing polarity.
4. **Set the supply voltage.** Disconnect sensors and Nano VIN. Power the converter from a current-limited supply and verify its output with a multimeter. Set the output between 5 V and 21 V according to the connected sensors. Switch power off before completing the wiring.
5. **Connect the Nano and INA219 supplies.** Apply the selected Nano power arrangement. Connect the INA219 VCC pins only to Nano 5 V and their GND pins to common GND. Configure the four unique I²C addresses.
6. **Insert the terminal carriers.** Check the terminal-to-board wall clearance.
7. **Connect the loops.** Route L+ from the converter to every sensor terminal. Route each L− individually through its assigned INA219 shunt to common GND, observing the VIN− / VIN+ polarity of the supplied schematic.
8. **Route and insulate the wires.** Use heat-shrink tubing on soldered joints; prevent strain at the Nano headers and avoid pinching wires under the lid.
9. **Install the cover and latch pin.** Check DIN-rail attachment and release.

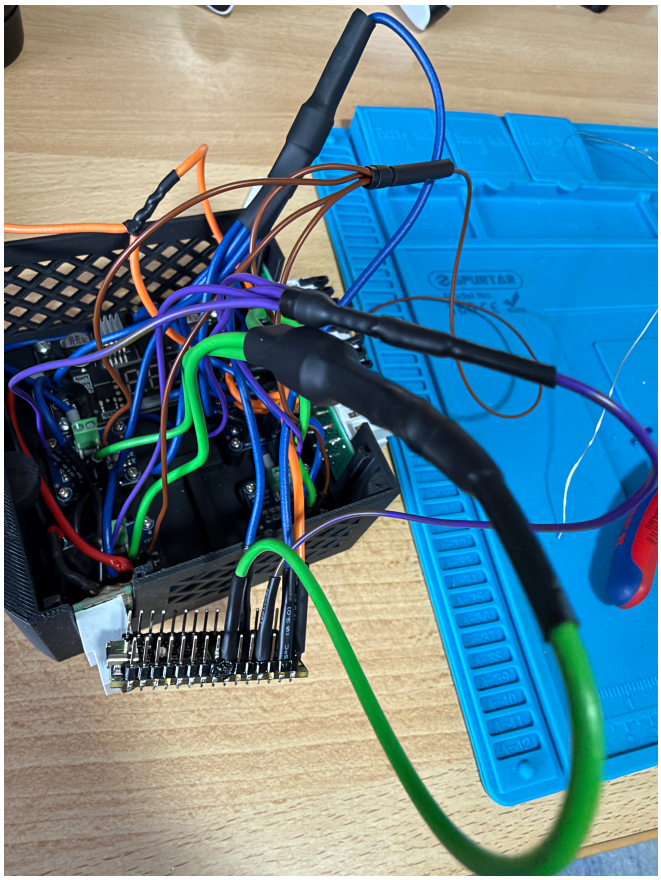


Figure 10: Prototype with the Nano header wiring and insulated shared logic connections accessible.

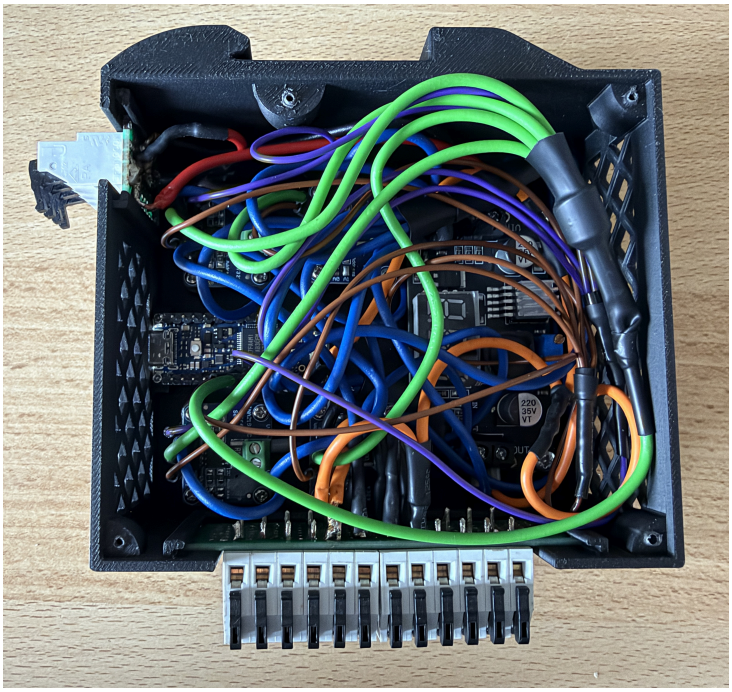


Figure 11: Assembled ai-4x-4to20mA prototype with the cover removed.

10 Checks before commissioning

USB messages are newline-terminated at 9600 baud. Initialize with `serveID`, `initStart`, then `initEnd`; the card reports its ID and `noSettings`. After INA219 setup, request index `0`–`3` for one integer current value. The ID prefix is `ai-4x-4to20mA_`; a missing INA219 causes an error state.

1. **Inspect the wiring and solder joints.** Check polarity, unintended bridges, exposed conductors and strain on the headers.
2. **Check continuity with power off.** Confirm bus poles 1/3 are joined for 24 V and poles 2/4 for 0 V, with no short between the rails. Check common ground and sensor L+ distribution. Each L− must reach GND only through its own shunt. A 0.1 Ω shunt can trigger a continuity beeper; visually inspect the path rather than relying on the beeper to exclude a bypass.
3. **Verify the supply arrangement.** Check the selected sensor voltage against the exact sensor specification and confirm the selected converter output between 5 V and 21 V.
4. **Check INA219 logic.** Confirm approximately 5 V at VCC, correct GND and four distinct I²C addresses: 0x40, 0x41, 0x44 and 0x45.
5. **Test the converter without sensors.** Confirm the selected voltage remains stable under the intended load and check component temperatures.
6. **Test one loop.** Connect a compatible two-wire sensor or loop-current test arrangement, with polarity checked. Verify 4, 12 and 20 mA against a reference meter; expected USB replies are approximately 4000, 12000 and 20000 µA.
7. **Test acquisition over USB.** Upload the matching firmware with `Adafruit_INA219` installed. Complete `serveID`, `initStart`, `initEnd`, then request indexes 0–3. Check missing-device errors and channel responses.
8. **Check Pi-side conversion.** For the AS007 with water, a received value of 12000 corresponds to 16 l/min using `(x - 4000) / 500`.
9. **Repeat for all channels.** Confirm enclosure channels 1–4 correspond to addresses 0x40, 0x41, 0x45 and 0x44 in that order. Test one loop at a time and verify that other channels remain independent.
10. **Check the mechanics.** Verify terminal retention, USB clearance, cover fit and secure DIN-rail mounting.

11 Finished module

The connector labels can be printed from the file `../docs/connector_labels.pdf`.

To attach them to the enclosure, first cover both sides of the printed paper with clear adhesive tape. This helps protect the labels from water and dirt. Then attach the labels to the measurement module using double-sided adhesive tape.

The following three images show the finished enclosure and label positions. Use L+ and L− labels for every channel according to the connection table. Use 24 V / 0 V labels for the bus input and pass-through.



Figure 12: Bottom view with ventilation grid.



Figure 13: Top view with the 24 V bus terminal and the USB-C socket.



Figure 14: Side view with the four L+ / L− sensor connections.